

**YEAR 10 SCIENCE EXTENSION
PHYSICS
ASSIGNMENT 3**

NAME: SOLUTIONSMARK: 42

1. A car engine supplies 4.50×10^3 N of force through the front wheels of a car and accelerates it uniformly from rest. The mass of the car is 1.40×10^3 kg.

- (a) Calculate the acceleration of the car.

Take forwards as +ve.

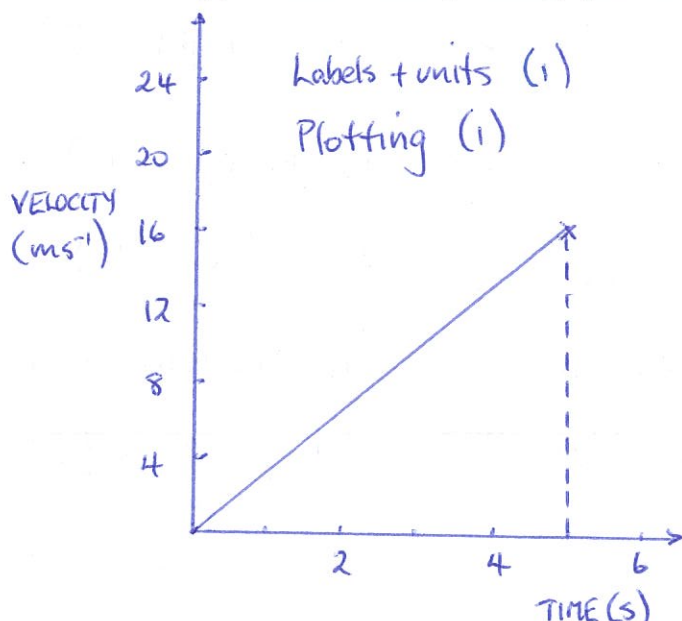
$$\begin{aligned}
 F &= 4.50 \times 10^3 \text{ N} \\
 m &= 1.40 \times 10^3 \text{ kg} \\
 a &= ?
 \end{aligned}
 \quad \Rightarrow \quad
 \begin{aligned}
 F &= ma \\
 a &= \frac{F}{m} \quad (1) \\
 &= \frac{4.50 \times 10^3}{1.40 \times 10^3} \quad (1) \\
 &= \underline{3.21 \text{ ms}^{-2} \text{ forwards}} \quad (1)
 \end{aligned}
 \quad (3)$$

- (b) Using your answer from part (a), calculate the velocity of the car after 5.0 s.

Take forwards as +ve

$$\begin{aligned}
 V &= ? \\
 u &= 0.0 \text{ ms}^{-1} \\
 a &= 3.21 \text{ ms}^{-2} \\
 t &= 5.0 \text{ s} \\
 s &= ?
 \end{aligned}
 \quad \Rightarrow \quad
 \begin{aligned}
 V &= u + at \quad (1) \\
 &= 0 + (3.21)(5.0) \quad (1) \\
 &= \underline{16.0 \text{ ms}^{-1} \text{ forwards}} \quad (1)
 \end{aligned}
 \quad (3)$$

- (c) Draw a velocity - time graph of the motion for 5.0 s and determine the displacement of the car.



$$\begin{aligned}
 \text{Displacement} &= \text{area under graph} \\
 &= \frac{1}{2}(5.0)(16.0) \quad (1) \\
 &= 40.0 \text{ m} \\
 \therefore \underline{s} &= \underline{40.0 \text{ m forwards}} \quad (1)
 \end{aligned}
 \quad (4)$$

2. A truck of mass 24.0 tonnes is moving at 80.0 kmh^{-1} along Roe Highway when the driver notices a small car that pulled out from a petrol station 60 m ahead. He hit the brakes sharply and just manages to avoid hitting the back of the car. The truck ended up moving at 30.0 kmh^{-1} and took 4.50 s to achieve this.

Calculate the average force exerted by the brakes of the truck.

$$v = 30.0 \text{ kmh}^{-1} = 8.33 \text{ ms}^{-1}$$

$$u = 80.0 \text{ kmh}^{-1} = 22.2 \text{ ms}^{-1}$$

$$a = ?$$

$$t = 4.50 \text{ s}$$

$$s = ?$$

Take forwards as +ve.

$$a = \frac{v-u}{t}$$

$$= \frac{8.33 - 22.2}{4.50} \quad (1)$$

$$= -3.08 \text{ ms}^{-2}$$

$$F = ma = (24.0 \times 10^3)(-3.08) \quad (1) \quad \therefore \text{Deceleration} = 3.08 \text{ ms}^{-2} \text{ backwards.} \quad (1)$$

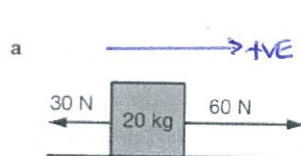
$$= 7.392 \times 10^4 \text{ N} \quad \therefore \text{Force} = 7.39 \times 10^4 \text{ N backwards.} \quad (1)$$

3. Explain why you "lurch forward" when standing on a bus or train and the vehicle brakes suddenly.

- You have inertia since you are moving. (1)
- As the vehicle slows, your inertia continues you forward with the initial speed of the vehicle. (1)

(2)

4. Calculate the nett force and the acceleration for the following objects. Show your calculations.



$$(a) \Sigma F = 60 - 30 = 30 \text{ N to the right.} \quad (1)$$

$$a = \frac{\Sigma F}{m} = \frac{30}{20} = 1.5 \text{ ms}^{-2} \text{ right.} \quad (1)$$

$$(b) \Sigma F = 250 - 100 = 150 \text{ N up} \quad (1)$$

$$a = \frac{\Sigma F}{m} = \frac{150}{75} = 2.0 \text{ ms}^{-2} \text{ up.} \quad (1)$$

$$(c) \Sigma F = 1800 - 150 = 1650 \text{ N forwards.} \quad (1)$$

$$a = \frac{\Sigma F}{m} = \frac{1650}{1500} = 1.1 \text{ ms}^{-2} \text{ forwards} \quad (1)$$

(6)

5. Which has the greater kinetic energy - a 40.0 g marble fired from a sling-shot at $1.50 \times 10^2 \text{ ms}^{-1}$ or a 40.0 kg child running at 1.50 ms^{-1} ? Justify your answer by calculation.

$$\begin{aligned} \text{Marble } E_k &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(0.0400)(1.50 \times 10^2)^2 \quad (1) \\ &= 4.50 \times 10^2 \text{ J} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{child } E_k &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(40.0)(1.50)^2 \\ &= 45.0 \text{ J} \quad (1) \end{aligned}$$

∴ Marble has the most E_k . (1)

(4)

6. A pile driver is used to push piles into the ocean bed to build a jetty. A $5.50 \times 10^2 \text{ kg}$ "driver" falls through 1.25 m to hit a pile.

- (a) How much potential energy does the "driver" have before it falls?

$$\begin{aligned} E_p &= mgh \\ &= (5.50 \times 10^2)(9.80)(1.25) \\ &= 6.738 \times 10^3 \text{ J} \\ \therefore \underline{E_p} &= \underline{6.74 \times 10^3 \text{ J}} \end{aligned}$$

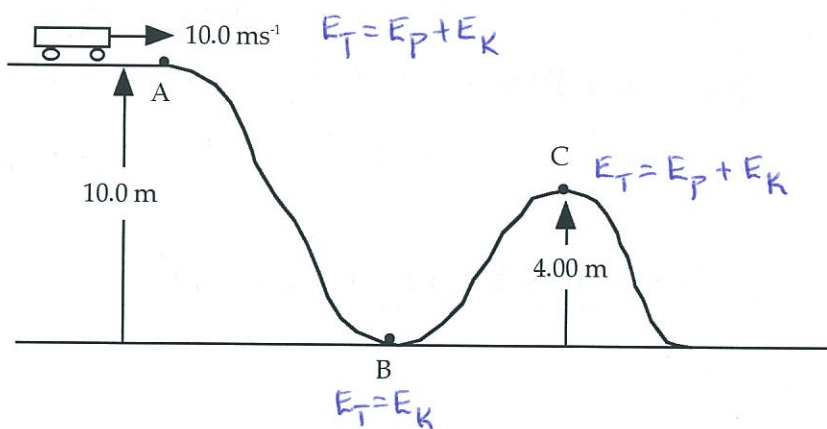
(3)

- (b) The potential energy becomes kinetic energy, and is a maximum as it hits the pile. Using this information, calculate the velocity of the driver as it hits.

$$\begin{aligned} E_T &= E_p(\text{top}) = E_k(\text{bottom}) \\ \Rightarrow mgh &= \frac{1}{2}mv^2 \\ \Rightarrow 6.738 \times 10^3 &= \frac{1}{2}(5.50 \times 10^2)v^2 \\ \Rightarrow v &= 4.95 \text{ ms}^{-1} \\ \therefore \underline{\text{velocity}} &= \underline{4.95 \text{ ms}^{-1} \text{ down.}} \end{aligned}$$

(3)

7. A 2.50×10^3 kg roller coaster is moving along a smooth track at 10.0 ms^{-1} as shown in the diagram below.



- (a) Calculate the total energy of the coaster at point A.

$$\begin{aligned}
 E_T &= E_P + E_K \\
 &= mgh + \frac{1}{2}mv^2 \quad (1) \\
 &= (2.50 \times 10^3)(9.80)(10.0) + \frac{1}{2}(2.50 \times 10^3)(10.0)^2 \quad (1) \\
 &= \underline{3.70 \times 10^5 \text{ J}} \quad (1)
 \end{aligned}$$

(3)

- (b) What would be its speed at point B?

$$\begin{aligned}
 E_T &= E_K \\
 &= \frac{1}{2}mv^2 \quad (1) \\
 \Rightarrow 3.70 \times 10^5 &= \frac{1}{2}(2.50 \times 10^3)v^2 \quad (1) \\
 \Rightarrow v &= 17.2 \text{ ms}^{-1} \\
 \therefore \underline{\text{speed} = 17.2 \text{ ms}^{-1}} \quad (1)
 \end{aligned}$$

(3)

- (c) How much kinetic energy does it still possess at point C?

$$\begin{aligned} E_T &= E_P + E_K \\ \Rightarrow E_T &= mgh + E_K \\ \Rightarrow E_K &= E_T - mgh \quad (1) \\ &= 3.70 \times 10^5 - (2.50 \times 10^3)(9.80)(4.00) \\ &= \underline{2.72 \times 10^5 \text{ J}}. \quad (1) \end{aligned}$$

(2)

- (d) In reality, your answers to (b) and (c) are incorrect. Explain why.

There is friction acting due to the air and contact with the rails. (1)

This results in energy being lost as heat and sound. (1)

(2)

